Please read the instructions below before the exam starts, and listen carefully to the instructions given by the invigilators.

You may read the question paper before the exam starts.

You must not write anything until the invigilator says that you may start.

You will be given five minutes after the exam has finished to complete the front of any answer books used.

May/June 2024

EC120 2023/4 A 800
EC120 2022/3 A 850

Formula sheet
Statistical tables

Only Casio FX-83GTPLUS or FX-85TGPLUS or FX-83GTx or FX-85GTx Casio FX- 85MS calculators are permitted

UNIVERSITY OF READING

INTRODUCTORY QUANTITATIVE METHODS IN ECONOMICS AND BUSINESS 2 AND STUDY SKILLS (EC120)

Two hours

This paper is divided into **TWO SECTIONS.**

Answer **ALL** questions in Section A (40 marks); and
Answer **TWO** questions in Section B (60 marks).

SECTION A
Answer ALL questions

1. In a dataset that contains students' test scores, one observation is identified as an outlier and as a possible influential observation. Discuss:

   (a) What is an influential observation?

   (3 marks)

   (b) What is an outlier, and how does it differ from an influential observation?

   (3 marks)

   (c) How can we identify an influential observation in a simple dataset in which we only have two variables? Suggest and compare two methods.

   (4 marks)

2. One large organisation, which employs 10,000 people, wants to compare wages of employees who declared a disability to wages of those who did not declare any disability. A simple random sample of 50 employees shows that the average wage among the 42 non-disabled employees is £21.32 per hour with a standard deviation of £5.29, while the average wage among the 8 disabled employees is £18.78 per hour with a standard deviation of £5.02.
   Based on these information, can we conclude that in this organisation there is a disability pay gap in favour of non-disabled employees? Discuss your answer in details.

   (10 marks)

3. Your friend is conducting a study to examine the relationship between the advertising expenditure and the sales revenue for a sample of companies in the retail sector. The regression analysis yields the following result:

   Sales Revenues = 5000 + 2.2 Advertising Expenditure

   (a) Interpret the slope coefficient in the context of this regression analysis.

   (3 marks)

   (b) Explain to your friend why they need to test the statistical significance of the slope coefficient and how this may affect the conclusion of their study.

   (7 marks)

4. Discuss:

   (a) What is a confidence interval and why is it useful?

   (6 marks)

   (b) How does the confidence interval change if we increase the size of our sample, for example from 100 to 1,000 (everything else equal)? (Hint: think about the formulas we use)

   (2 marks)

   (c) How does the confidence interval change if we increase the level of confidence, for example, from 95% to 99%? (Hint: think about the values in the table of the t-statistics)

   (2 marks)

SECTION B
Answer TWO questions
(30 marks each question)

5. For a sample of 8 regions, the table below gives the inflation rates and the level of consumer spending in 2004.

Region	Inflation rate	Consumer spending (Million £)
1	2.5	500
2	3.2	480
3	1.8	550
4	4.5	400
5	2.0	520
6	3.8	450
7	1.5	580
8	5.0	350

(a) Draw a scatterplot of these data. Describe the overall pattern in the scatterplot and any striking deviations from the pattern.

(5 marks)

(b) Compute the sample mean and standard deviation of the inflation rate and of consumer spending.

(6 marks)

(c) Compute (by hand) the correlation between self-esteem and average marks. For full marks, show the steps in your computation.

(5 marks)

(d) Find a regression line which allows you to predict inflation given information on consumer spending.

(5 marks)

(e) Interpret the results of this regression.

(3 marks)

(f) Discuss why we may not be able to interpret the regression results as the causal effect of consumer spending on inflation rate. (Hint: think about the assumptions needed for inference on the regression parameters)

(6 marks)

6. You have information on the average amount spent on transport and on the average amount spent on food of two samples of University of Reading students living in Accommodation A and in Accommodation B.

For the two simple random samples, the following table gives the averages, their standard deviations and the sample sizes.

	Accommodation A		Accommodation B	
	Mean	St. Dev.	Mean	St. Dev.
Spending on transport per week	23.42	5.12	33.45	4.77
Spending on food per week	53.12	10.90	49.21	8.36
Observations	25		30	

(a) Describe the appropriate null and alternative hypotheses for testing whether:

(i) spending on transport per week of students in Accommodation A is the same as spending on transport per week of students in Accommodation B;

(ii) spending on food per week of students in Accommodation A is the same as spending on food per week of students in Accommodation B.

(4 marks)

(b) Carry out significance tests (using a 5% level of significance) to test these hypotheses (also state which critical value you are using and why).

(5 marks)

(c) Compute the 95% confidence interval for the difference in means for each hypothesis.

(6 marks)

(d) Interpret the results of the confidence intervals. What do we conclude?

(4 marks)

(e) Compute the 95% confidence interval around the mean for spending on food per week of students in Accommodation B and discuss (without further computations) whether you would expect to reject the following Null Hypothesis or not at the 5% level of statistical significance:
H0: μ spending food B = 50
Ha: μ spending food B ≠ 50

(7 marks)

(f) List all the assumptions necessary for:

(i) the statistical inference you have carried out in point (e);

and for

(ii) the statistical inference you have carried out in point (b).

(4 marks)

7. A researcher is investigating the relationship between the amount of air pollution (measured in particulate matter concentration, PM2.5 in micrograms per cubic meter) and the health status of residents in a city. The sample consists of 150 residents randomly located around the city. Since she is uncertain what is the best model, she tries two models.

   The first simple linear regression equation for this relationship is:

   $$\widehat{Health\ Status} = 80 - 0.5PM$$

   St.err. of the slope coefficient: 0.03

   The second simple linear regression equation for this relationship is:

   $$\widehat{Log\ Health}\ Status = 1.90 - 0.11PM$$

   St.err. of the slope coefficient: 0.01

   (a) Interpret the results of slope in the two cases.

   (6 marks)

   (b) At the 5% level of statistical significance, test the hypothesis that PM has no effect on health for both equations. What do you conclude?

   (8 marks)

   (c) Discuss why the researcher may have decided to try two different versions: with and without logs. (Hint: think about the assumptions of this model)

   (8 marks)

   (d) How could the researcher check that one model is more appropriate than the other? Suggest two options.

   (8 marks)

(End of Question Paper)